

Research Focus

My research develops the foundations for evaluating, debugging, and iteratively improving agentic AI, including goal-oriented reliability evaluation, trace-level failure analysis, machine unlearning, causal reasoning for root-cause analysis, and trustworthy data pipelines for agent training and evaluation.

Technical Expertise

Research Areas	Agentic AI, Self-Improving Multi-Agent Systems, AI Reliability & Evaluation, LLM Agents, Causal Inference, Machine Unlearning, Data-Centric AI, Explainable AI, Responsible AI
Systems & Tools	Python, PyTorch, TensorFlow, LangChain, LangGraph, Vector Databases, AWS, Azure, GCP, Distributed ML Systems, LLM Evaluation Frameworks, Agent Trace Analysis Pipelines
Applications	Agent Reliability & Observability, Root-Cause Analysis, Predictive Diagnostics, Enterprise AI, Trustworthy NL2SQL, Retrieval-Augmented Generation

Academic Experience

2023 - Present	Assistant Professor , Department of Computer Science Field Membership: Computer Science, Statistics and Data Science Cornell University
2021 - 2023	Computing Innovation Fellow Postdoctoral Researcher , Department of Computer Science The University of Chicago Advisor: Prof. Raul Castro Fernandez

Education

2017 - 2021	PhD in Computer Science , University of Massachusetts Amherst, USA Advisor: Prof. Barna Saha Thesis title: Robust Algorithms for Clustering with Applications to Data Integration
2015 - 2017	Masters in Computer Science , University of Massachusetts Amherst, USA
2010 - 2014	BTech in Computer Science and Engineering , Indian Institute of Technology (IIT) Delhi, India Advisor: Prof. Amitabha Bagchi

Selected Publications

Selected from 50+ peer-reviewed publications. Full list under “Refereed Conference and Journal Publications.”

- [1] Vishwanath Katharki, **Sainyam Galhotra**. Goal-Oriented Reliability Evaluation for Multi-Agent Systems. ACM Conference on AI and Agentic Systems (**CAIS**), 2026.
- [2] Anna Mazhar, Huzaifa Suri, **Sainyam Galhotra**. Trace-Level Analysis of Information Contamination in Multi-Agent Systems. ACM Conference on AI and Agentic Systems (**CAIS**), 2026.
- [3] Eric Enouen, **Sainyam Galhotra**, et al. Debugging Concept Bottleneck Models through Removal and Retraining. International Conference on Learning Representations (**ICLR**), 2026.
- [4] Samyak Jain, Ronak Kalvani, **Sainyam Galhotra**, Sayan Ranu. Is Graph Unlearning Ready for Practice? A Benchmark on Efficiency, Utility, and Forgetting. **ICLR**, 2026.
- [5] Steven Xia, Minbiao Han, Jonathan Light, Raul Castro Fernandez, Haifeng Xu, **Sainyam Galhotra**. Optimal Pricing for Data-Augmented AutoML Marketplaces. International Conference on Machine Learning (**ICML**), 2026.

- [6] Shubham Mohole, **Sainyam Galhotra**. VeriMinder: Mitigating Analytical Vulnerabilities in NL2SQL. **ACL**, 2025.
- [7] **Sainyam Galhotra**, Joseph Y. Halpern. Intervention and Conditioning in Causal Bayesian Networks. **NeurIPS**, 2024.

Selected Awards and Honors

- 2025 Distinguished Reviewer Award at SIGMOD
- 2021-2023 Computing Innovation Fellowship Award Recipient (by CRA, CCC and NSF)
- 2023 Most reproducible paper award (Honorable Mention) in SIGMOD
- 2021 Rising Star in Data Science at the Data Science Institute, UChicago
- 2021 DAAD AInet Fellow
- 2020 ACM SIGMOD Entity Resolution Programming Contest – Top 5 finalist
- 2018 - 2019 Most reproducible paper award in SIGMOD 2018 and 2019
- 2017 Best paper award in SIGSOFT FSE 2017
- 2016 First recipient of Krithi Ramamritham Computer Science Scholarship
- 2010 All India Rank 53 in IIT-JEE (of $\sim 450K$ candidates)
- 2010 State Rank 1 and All India Rank 20 in AIEEE (of $\sim 1M$ candidates)

Funding

- US Army Research Office (**ARO**) Generating Reliable, Verifiable, and Contextualized Explanations (jointly with Joseph Y. Halpern)
Duration: Sept, 2025 - Aug, 2028
Amount: \$727,399
- United States - Israel Binational Science Foundation (**BSF**) Data-Driven Responsible Causal Conclusions (jointly with Brit Youngmann, faculty at Technion)
Duration: Sept, 2025 - Aug, 2027
Amount: \$180,000
- LinkedIn** PhD Fellowship (awarded to Eric Enouen)
Duration: Sept, 2025 - Aug 2026
Amount: \$70,000
- CCSS** Cornell Center for Social Sciences
Duration: Sept, 2025 - Aug 2026
Amount: \$10,000
- Infosys** Data discovery for complex ML tasks
Duration: Sept, 2024 - Jun, 2025
Amount: \$100,000, as part of \$950,000 joint award
- LinkedIn** Responsible Data Science (Amount: \$1,000)
- Microsoft** Microsoft Azure Cloud Computing Grant
Duration: Oct 6, 2023 - Jun 30, 2024
Amount: \$10,000
- NSF** Designing a Privacy-aware Fair Trade Marketplace for Data
The National Science Foundation CRA Computing Innovation Fellowship
Duration: Sept 1, 2021 – July 14, 2023
Amount: \$267,500

Work Experience

- Aug - Nov 2019 **Applied Scientist Intern**, Amazon, Seattle, USA
Developed a novel deep-learning based entity linking approach that leveraged Amazon Knowledge Graph to identify product attributes from search queries.

- May - Aug 2019 **Research Intern**, IBM T.J. Watson Research Center, Yorktown Heights, USA
Demonstrated use of external knowledge for feature enrichment with AutoML. Demo paper of this work was presented at CIKM 2020 and ICDM 2019.
- May - Aug 2018 **Research Intern**, Megagon Labs, Mountain View, USA
Designed a scalable system to generate labelled training data with minimum human intervention. Demonstrated superior performance over a variety of NLP tasks. This work was published at SIGMOD 2021.
- May - Aug 2016 **Research Intern**, Google Research, Mountain View, USA
Developed algorithms to recommend feature engineering services for billions of internal datasets
- May 2014 - 2015 **Predoctoral Budding Scientist**, Xerox Research Center India (XRCI), Bangalore India
Worked on data mining algorithms to reduce customer churn and validated it on various telecommunications providers.

Books

- [1] Ziawasch Abedjan, Mahdi Esmailoghli, **Sainyam Galhotra**. Data Discovery in Data Lakes, Springer Nature.

Selected Research Projects

Reliable Agentic AI and Self-Improving Multi-Agent Systems. Designing infrastructure for reliable AI agents through goal-oriented evaluation, trace-level monitoring, automated failure analysis, and iterative self-improvement. Focus areas include agent reliability, evaluation pipelines, synthetic data generation, workflow optimization, and multi-agent coordination.

Machine Unlearning Reliability and Evaluation. Developing principled frameworks to test and validate machine unlearning systems with reliability guarantees, robustness analysis, and adversarial evaluation. Extending to unlearning of AI agents.

Multi-Agent Information Contamination Analysis. Building trace-level methods for identifying information corruption, hallucination propagation, and reasoning failures across collaborating AI agents and tool-use pipelines.

Data Discovery and Data-Centric AI. Developing scalable systems for discovering, profiling, integrating, and curating data for machine learning and causal inference. Focus on trustworthy data acquisition, representative data selection, and explainable data preparation.

Causal AI for Real-World Decision-Making. Building multimodal causal reasoning systems for root-cause analysis, predictive diagnostics, and prescriptive decision-making in industrial settings. Combines knowledge graphs, sensor streams, image data, and explainable AI workflows for real-time operational intelligence.

Refereed Conference and Journal Publications

Articles published at conferences

- [1] Eric Enouen, **Sainyam Galhotra**, et al. Debugging Concept Bottleneck Models through Removal and Retraining. International Conference on Learning Representations (**ICLR**), 2026.
- [2] Rahul Raychaudhury, Aryan Esmailpour, **Sainyam Galhotra**, Stavros Sintos. Metric k-Clustering using only Weak Comparison Oracles. International Conference on Learning Representations (**ICLR**), 2026.
- [3] Samyak Jain, Ronak Kalvani, **Sainyam Galhotra**, Sayan Ranu. Is Graph Unlearning Ready for Practice? A Benchmark on Efficiency, Utility, and Forgetting. International Conference on Learning Representations (**ICLR**), 2026.
- [4] Vishwanath Katharki, **Sainyam Galhotra**. Goal-Oriented Reliability Evaluation for Multi-Agent Systems. ACM Conference on AI and Agentic Systems (**CAIS**), 2026.

- [5] Anna Mazhar, Huzaifa Suri, **Sainyam Galhotra**. Trace-Level Analysis of Information Contamination in Multi-Agent Systems. ACM Conference on AI and Agentic Systems (**CAIS**), 2026.
- [6] Eugenie Y. Lai, Inbal Croitoru, Brit Youngmann, **Sainyam Galhotra**, El Kindi Rezig, Michael J. Cafarella. Interpretable Attribute Discretization. In **SIGMOD**, 2026.
- [7] Steven Xia, Minbiao Han, Jonathan Light, Raul Castro Fernandez, Haifeng Xu, **Sainyam Galhotra**. Optimal Pricing for Data-Augmented AutoML Marketplaces. International Conference on Machine Learning (**ICML**), 2026.
- [8] Benton Li, Nativ Levy, Brit Youngmann, **Sainyam Galhotra**, Sudeepa Roy. Fair and Actionable Causal Prescription Ruleset. In **SIGMOD**, 2025.
- [9] Eugenie Y. Lai, Inbal Croitoru, Noam Bitton, Ariel Shalem, Brit Youngmann, **Sainyam Galhotra**, El Kindi Rezig, Michael J. Cafarella. SeerCuts: Explainable Attribute Discretization. In **SIGMOD**, 2025.
- [10] Vivek Vishwakarma, **Sainyam Galhotra**. Responsible defect detection with probabilistic contrastive counterfactuals. In **SPIE**, 2025.
- [11] Shubham Mohole, Hongjun Choi, Shusen Liu, Christine Klymko, Shashank Kushwaha, Derek Shi, Wesam Sakla, **Sainyam Galhotra**, Ruben Glatt. VERIRAG: Healthcare Claim Verification via Statistical Audit in Retrieval-Augmented Generation. In **BCB**, 2025.
- [12] Shubham Mohole, **Sainyam Galhotra**. VeriMinder: Mitigating Analytical Vulnerabilities in NL2SQL. In **ACL**, 2025.
- [13] Shubham Mohole, **Sainyam Galhotra**. FAIR-SHEPHERD: An Interactive System for Auditable Fairness via Population-Grounded Surrogates. In **ICDM**, 2025.
- [14] Rahul Raychaudhury, Wen-Zhi Li, Syamantak Das, **Sainyam Galhotra**, Stavros Sintos. Metric Clustering and Graph Optimization Problems using Weak Comparison Oracles, (**COLT**), 2025.
- [15] Aryan Esmailpour, **Sainyam Galhotra**, Rahul RayChaudhary and Stavros Sintos. A theoretical framework for distribution-aware dataset search. In **PODS**, 2025.
- [16] **Sainyam Galhotra**, Rahul RayChaudhary and Stavros Sintos. Efficient Algorithms for k-Clustering with Noisy and Exact Oracles. In **PODS**, 2025.
- [17] Andrew Kang, and **Sainyam Galhotra**. Type-driven data discovery. In IEEE **BigData** undergraduate symposium, 2024.
- [18] **Sainyam Galhotra**, and Joseph Y. Halpern. Intervention and Conditioning in Causal Bayesian Networks. In Conference of Neural Information Processing Systems. In **NeurIPS**, 2024.
- [19] Lennart Behme, **Sainyam Galhotra**, Kaustubh Beedkar, Volker Markl. Fainder: A Fast and Accurate Index for Distribution-Aware Dataset Search. In **PVLDB** 2024.
- [20] Yash Kurkure, Miles Shamo, Joseph Wiseman, **Sainyam Galhotra**, and Stavros Sintos. Faster Algorithms for Fair Max-Min Diversification in R^d . In **SIGMOD**, 2024.
- [21] Yue Gong, **Sainyam Galhotra**, and Raul Castro Fernandez. Nexus: Correlation Discovery over Collections of Spatio-Temporal Tabular Data. In **SIGMOD**, 2024.
- [22] Kevin Dharmawan, Chirag A Kawediya, Yue Gong, Zaki Indra Yudhistira, Zhiru Zhu, **Sainyam Galhotra**, Adila Krisnadhi, Raul Castro Fernandez. Demonstration of Ver: View Discovery in the Wild. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), 2024.
- [23] Jiongli Zhu, **Sainyam Galhotra**, Nazanin Sabri, and Babak Salimi. Crab: Learning Certifiably Fair Predictive Models in the Presence of Selection Bias. (**PVLDB**) 2023.
- [24] **Sainyam Galhotra**, Yue Gong and Raul Castro Fernandez. METAM: Goal-Oriented Automated Data Discovery. IEEE International Conference on Data Engineering (**ICDE**) 2023.
- [25] Yue Gong, Zhiru Zhu, **Sainyam Galhotra**, and Raul Castro Fernandez. Ver: View Discovery in the Wild. IEEE International Conference on Data Engineering (**ICDE**) 2023.
- [26] Fangzhu Shen, Kayvon Heravi, Oscar Gomez, **Sainyam Galhotra**, Amir Gilad, Sudeepa Roy, and Babak Salimi. Causal What-If and How-To Analysis Using Hyper. IEEE International Conference on Data Engineering (**ICDE**) 2023.

- [27] **Sainyam Galhotra**, Donatella Firmani, Barna Saha and Divesh Srivastava. Hierarchical Entity Resolution using an Oracle. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), 2022.
- [28] **Sainyam Galhotra***, Amir Gilad*, Sudeepa Roy, and Babak Salimi. Hyper: Hypothetical Reasoning With What-If and How-To Queries Using a Probabilistic Causal Approach. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), 2022.
Most Reproducible Paper Award Honorable Mention
- [29] **Sainyam Galhotra**, Anna Fariha, Raoni Lourenco, Juliana Freire, Alexandra Meliou and Divesh Srivastava. DataPrism: Exposing Disconnect between Data and Systems. In **SIGMOD**, 2022.
- [30] **Sainyam Galhotra**, Karthikeyan Shanmugam, Prasanna Sattigeri and Kush R. Varshney. Causal Feature Selection for Algorithmic Fairness. In **SIGMOD**, 2022.
- [31] Suman Bera, Syamantak Das, **Sainyam Galhotra** and Sagar Kale. Fair k-Center in MPC and Streaming Settings. The WebConf **WWW**, 2022.
- [32] **Sainyam Galhotra**, Sandhya Saisubramanian and Shlomo Zilberstein. Learning to Generate Fair Clusters from Demonstrations. In Proc. of 4th ACM Conference on AI, Ethics, and Society (**AIES**), 2021.
- [33] **Sainyam Galhotra**, Romila Pradhan and Babak Salimi. Explaining Black-Box Algorithms Using Probabilistic Contrastive Counterfactuals. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), 2021.
- [34] **Sainyam Galhotra**, Donatella Firmani, Barna Saha and Divesh Srivastava. BEER: Blocking for Effective Entity Resolution. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), Demonstrations Track, 2021.
- [35] Paul Y Wang, **Sainyam Galhotra**, Romila Pradhan and Babak Salimi. Demonstration of Generating Explanations for Black-Box Algorithms Using Lewis. In **PVLDB**, Demonstrations Track, 2021.
- [36] Udayan Khurana and **Sainyam Galhotra**. Semantic Concept Annotation for Tabular Data. In Proc. of ACM International Conference on Information and Knowledge Management (**CIKM**), 2021.
- [37] **Sainyam Galhotra**, Behzad Golshan and Wang-Chiew Tan. Adaptive Rule Discovery for Labeling Text Data. In **SIGMOD**, Data Science and Engineering Track, 2021.
- [38] Raghavendra Addanki, **Sainyam Galhotra**, and Barna Saha. How to Design Robust Algorithms using Noisy Comparison Oracle. In **PVLDB**, 2021.
- [39] **Sainyam Galhotra** and Udayan Khurana. Semantic Search over structured data. In Proc. of Conference on Information and Knowledge Management (**CIKM**), Demonstrations Track, 2020.
Nomination: Best Demo Paper
- [40] Sandhya Saisubramanian*, **Sainyam Galhotra*** and Shlomo Zilberstein. Balancing the Tradeoff Between Clustering Value and Interpretability. In **AIES**, 2020.
- [41] **Sainyam Galhotra**, Udayan Khurana, Oktie Hassanzadeh, Kavitha Srinivas, Horst Samulowitz, Miao Qi. Automated Feature Enhancement for Predictive Modeling using External Knowledge. In **ICDM**, Demonstrations Track, 2019.
- [42] **Sainyam Galhotra**, Soumyabrata Pal, Arya Mazumdar and Barna Saha. Connectivity in Random Annulus Graphs and the Geometric Block Model. In Proc. of International Conference on Randomization and Computation (**APPROX-RANDOM**), 2019.
- [43] **Sainyam Galhotra**, Donatella Firmani, Barna Saha and Divesh Srivastava. Robust Entity Resolution using Random Graphs. In **SIGMOD**, 2018.
Most Reproducible Paper Award
- [44] **Sainyam Galhotra**, Arya Mazumdar, Soumyabrata Pal and Barna Saha. The Geometric Block Model. In Proc. of the 32nd AAAI Conference on Artificial Intelligence (**AAAI**), 2018.
- [45] **Sainyam Galhotra**, Arya Mazumdar, Soumyabrata Pal and Barna Saha. The Geometric Block Model and Applications. In Proc. of Allerton Conference on Communication, Control, and Computing (Invited papers track), pages 1147-1150, 2018.

- [46] **Sainyam Galhotra**, Yuriy Brun and Alexandra Meliou. Fairness Testing: Testing Software for Discrimination. In (ESEC/SIGSOFT FSE), 2017.
Best Paper Award
- [47] Akhil Arora*, **Sainyam Galhotra*** and Sayan Ranu. Debunking the Myths of Influence Maximization. In Proc. of ACM International Conference on Management of Data (**SIGMOD**), 2017.
Most Reproducible Paper Award
- [48] **Sainyam Galhotra***, Akhil Arora* and Shourya Roy. Holistic IM: Combining Scalability and Efficiency with Opinion-Aware Models. In **SIGMOD**, 2016
- [49] Shourya Roy, Sandipan Dandapat, R. Mariappan, S Srivastava, **Sainyam Galhotra** and B. Peddamuthu. QA^{RT} : A System for Real-time Holistic Quality Assurance for Contact Center Dialogues. In **AAAI**, 2016.
- [50] **Sainyam Galhotra**, Amitabha Bagchi, Srikanta Bedathur, Maya Ramanath and Vidit Jain. Tracking the Conductance of Rapidly Evolving Topic-Subgraphs. In **VLDB**, 2015.
- [51] **Sainyam Galhotra**, Akhil Arora, Srinivas Virinchi and Shourya Roy. ASIM: A Scalable Algorithm for Influence Maximization under the Independent Cascade Model. In Proc. of ACM International Conference on World Wide Web (**WWW**), 2015 (Poster: Companion Volume)
- [52] STAR: Real-time Spatio-Temporal Analysis and Prediction of Traffic Insights using Social Media. In **CoDS-COMAD**, 2015.
- [53] Vidit Jain and **Sainyam Galhotra**. Min-d-Occur: Ensuring Future Occurrences in Streaming Sets. In Proc. of the Conference on Uncertainty in Artificial Intelligence (**UAI**), 2014.

Tutorials

- [1] Ziawasch Abedjan, Mahdi Esmailoghli, **Sainyam Galhotra**. Data Discovery in Data Lakes: Operations, Indexes, Systems. VLDB 2025.
- [2] Daniel Alabi, **Sainyam Galhotra**, Shagufta Mehnaz, Zeyu Song, Eugene Wu. Privacy and Security in Distributed Data Markets. SIGMOD 2025.
- [3] Romila Pradhan, Aditya Lahiri, **Sainyam Galhotra** and Babak Salimi. Explainable AI: Foundations, Applications, Opportunities for Data Management Research. SIGMOD 2022.
- [4] Romila Pradhan, Aditya Lahiri, **Sainyam Galhotra** and Babak Salimi. Explainable AI: Foundations, Applications, Opportunities for Data Management Research. ICDE 2022.
- [5] Akhil Arora*, **Sainyam Galhotra*** and Sayan Ranu. Navigating the Maze of Influence Maximization Algorithms. IEEE DSAA, 2019.
- [6] Akhil Arora*, **Sainyam Galhotra*** and Sayan Ranu. Influence Maximization Revisited: The State of the Art and the Gaps that Remain. In Proc. of EDBT, 2019.

Teaching Experience

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|------------------------|--|
| Fall 2024, Fall 2025 | Introduction to Database Systems (CS 4320), Cornell University
Total Number of students: 719 |
| Fall 2023, Spring 2025 | Data to Decisions: Principles of Efficient Data Science (CS 6386), Cornell University
Total Number of students: 58 |
| Summer 2024 | Instructor at CSMore program to broaden the participation of undergraduates in computer science through interactive learning, research exposure, professional development, and social engagement. |
| Dec 2022 | Guest Lecture on Data Science: University of California San Diego for high-school students (Morse High School), Delhi Public School, UAE and Harvard Extension - Data Mining for Business |

Mentorship

PhD Students:

- Anna Mazhar

- Eric Enouen
- Ning Wang
- Wenzhi Li

External Students:

- Rahul Raychaudhury (PhD student, Duke University): Paper published at PODS 2025, COLT 2025
- Eugenie Lai (PhD student, MIT): Paper published at SIGMOD 2025
- Lennart Behme (PhD student, TU Berlin): Paper published at PVLDB 2024
- Yash Kurkure (UIC): Paper published at SIGMOD 2024
- Miles Shamo (UIC): Paper published at SIGMOD 2024
- Joseph Wiseman (UIC): Paper published at SIGMOD 2024

Undergraduate/M.Eng. Students:

- Shubham Mohole (2024): Paper published at ACL, ICDM and BCB 2025
- Benton Li (2023): Paper published at SIGMOD 2025
- Stanley Jiang (2023-2024)
- Bryan Liang (2023)
- Andrew Kang (2024)
- Brandon Li (2024)
- Ved Sriram (2024)
- Anthony Song (2024)

As a postdoctoral researcher, I mentored two PhD students for their research projects and three Masters students for semester projects at UChicago. Further, I mentored four Masters and PhD students at UCSD and Duke University as a part of my external collaborations.

- Yue Gong (PhD student, UChicago 2022)
- Zhiru Zhu (PhD student, UChicago 2022)
- Jiongli Zhu (PhD student, UCSD 2022)
- Paul Wang (Masters student, UCSD 2020)

Professional Service

- 2026 Submission Chair: DASHSys 2026 – Workshop on Systems for Data-centric Agents with Human-in-the-loop, co-located with VLDB 2026
- 2024 Initiated the Mentorship program for the Database community
First edition was at SIGMOD 2024, Next edition at EDBT 2026.
- 2024, 2025 Co-chair: Guide-AI workshop at SIGMOD
- 2023-2025 Co-chair: Tabular Data Analysis workshop at VLDB
- 2024 Co-chair: New Researcher Symposium at SIGMOD
- 2023-present Served as the faculty head for Computational and Quantitative Social Sciences
Panelist: ICO/Alan Turing Institute workshop on Fairness in AI
NSF Review Panels : SBIR/STTR 2022, CSGrad4US 2022
Senior PC Member: IJCAI (2021)
PC Member: SIGMOD (2024, 2025, 2026), ICDM (2024), CIDR (2026), ICDE (2023), VLDB (2024, 2025), UAI (2021), AAAI (2022, 2021, 2020), FAccT (2021), SIGMOD Reproducibility 2018,
Reviewer: FOCS (2021), AISTATS (2021), ICDE (2021) NeurIPS (2021, 2020, 2019, 2018, 2016), ICML (2020,2019), NORDICHI 2020, CIKM (2020, 2019, 2018, 2017), KDD (2018, 2017), SIGMOD 2018, TKDE, SEA 2017, WWW 2017, CoDS 2015
- 2019-Present **Feature Editor** for the ACM XRDS Magazine.